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Respondent

75

Anonymous

1412:59

Time to complete

A. General information about the respondent

1. * Country

Mongolia



2. * Name of organization(s) or entity(s) responsible for the preparation of the report

Ministry of Economy and Development

3. * Website of organization(s) or entity(s) responsible for the preparation of the report

https://med.gov.mn/

4. * Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

6. Please describe the role/mandate of your organization

The role of the Ministry is to improve the social and economic potential of Mongolia in a sustainable and accessible way by providing methodology and management for developing, planning and implementing long-, medium- and short-term development policies of Mongolia integrated

7. * Full name of officially designated contact person(s) who completed this survey

Munkhtuya Lkhamsuren

8. * Position of officially designated contact person(s) who completed this survey

Senior officer, Department of Integrated Science and Technology Policy, Ministry of Economy and Development

9. * Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

11. Brief description of the consultation process established for the preparation of the report

12. * Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

13. * Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey
(Please use commas to separate multiple links, as: link A, link B)

14. * Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

☐ Public

☐ Private

☐ Civil Society

☐ Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

☒ Yes

☐ No

17. 1.1 cont. **If yes**, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

Ministry of Digital development, Innovation and communications, State-owned research institutions, State-owned universities

18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

☒ Yes

☐ No

19. 1.2 cont. **If yes**, which language(s)?

Mongolian

20. 1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: (i) 16.f, 16.g, 16.h)

[i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

☒ Yes

☐ No

21. 1.3 cont. **If yes**, please provide details on the activities organized or foreseen and links as relevant.

1. Open Access Seminar in Mongolia (2025) Organized by EIFL (Electronic Information for Libraries), this seminar focuses on: • Publishing open access journals and books. • Developing open access repositories. • Promoting the adoption of open access policies. • Strategies for collective advocacy of open access to research results. The event aims to enhance understanding of open access publishing and its benefits, fostering a culture of openness in Mongolia's academic community. 2. National Training Workshop on Science, Technology, and Innovation Parks (April 2025) Co-organized by the Ministry of Economy and Development and UNCTAD, this workshop focuses on: • Designing, operating, and managing STI parks. • Enhancing STI capacity to promote sustainable development. • Empowering women and youth entrepreneurs in STI. The workshop aims to strengthen Mongolia's capacity to implement STI policies and enhance the country's STI capacity, ultimately promoting sustainable development. 3. High-Performance Supercomputing Center Project The Ministry of Economy and Development, in collaboration with the Mongolian Academy of Sciences and other institutions, is working on establishing a High-Performance Supercomputing Center. This project aims to: • Enhance data analysis capabilities in various sectors. • Support research in artificial intelligence and machine learning. • Provide access to advanced research technology for scientists and students. MongolBeat The project is expected to elevate the development of science, technology, and innovation in Mongolia. 4. National Conference of Mongolian Young Scientists (2024) Held under the theme "Open Science – Youth Participation," this conference focused on: • Conceptual renewal of science in Mongolia. • Development and financing of human resources. • Legalizing cooperation between state and private sectors. The conference aimed to engage young scientists in discussions on open science and its implementation in Mongolia. 5. International Collaboration and Conferences Mongolian scientists have participated in various international conferences, such as: • "Youth Science 2024" International XI Conference in Belarus. • "Geology and Exploration of Mongolia-2025" Conference. These conferences provided platforms for open dialogue and engagement with other knowledge systems, fostering international collaboration in research.

22. 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[i], in publicly funded research? (ref: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the guiding principles for open science, namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

☒ Yes

☐ No

23. 1.4 cont. If **yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

Specific actions have been undertaken and are planned to be implemented in Mongolia by the end of 2025 to incorporate the values and guiding principles of open science into publicly funded research, in alignment with the 2021 UNESCO Recommendation (sections 16.a–16.e, 16.h). The Ministry of Economy and Development, together with the Mongolian Academy of Sciences (MAS), has initiated a review and revision of national science and innovation policies to reflect key open science principles such as quality, integrity, equity, transparency, and inclusiveness. These values are expected to be embedded in the forthcoming National Science, Technology, and Innovation Roadmap (2025–2035). At the institutional level, universities and public research organizations—such as the National University of Mongolia, the University of Finance and Economics (UFE), and the Information Technology Center of MAS—have developed and are maintaining institutional repositories to facilitate open access to scientific publications and research data. These platforms enhance transparency, data reproducibility, and collective benefit by making the results of publicly funded research openly available to the broader community. Moreover, national research funding mechanisms are increasingly designed to promote inclusivity and equitable access. Funding calls now prioritize the participation of young researchers, women scientists, and institutions in rural or less-resourced areas, and encourage collaboration between public and private sectors. Efforts are also being made to integrate traditional knowledge systems into research projects, reinforcing values of diversity and respect. Looking ahead, Mongolia is working toward the establishment of a national open science framework that will provide formal guidelines on responsible data management, open access, open-source software use, and research transparency. This framework aims to institutionalize open science practices across all stages of the research cycle, ensuring flexibility, accountability, and sustainability in publicly funded research.

Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

24. 2.1 Does your country have a national policy[i], strategy or plan of action on science, technology and innovation (STI)?

[i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

☐ Yes

☒ No

25. 2.1 cont. If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Mongolia does not currently have a formal, comprehensive national policy, strategy, or plan of action specifically dedicated to science, technology, and innovation (STI). However, the government and relevant ministries are actively working toward developing such a framework to better coordinate and support scientific research, technology development, and innovation across the country. Efforts are underway to consult stakeholders and align future policy initiatives with international standards and recommendations, including the principles outlined in the 2021 UNESCO Recommendation on Open Science.

26. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

- ☒ Yes
- ☐ Partly
- ☐ No

27. 2.2 cont. **If yes or partly**, please share a machine-readable PDF file(s) of the policy(ies) using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

Mongolia has a national legal framework that addresses open science and its key elements in alignment with the values and principles outlined in the 2021 UNESCO Recommendation on Open Science. Specifically, the Law on Science and Technology, which was renewed in 2024, provides the foundation for governing science, technology, and innovation activities in the country. This law incorporates provisions promoting transparency, open access to scientific knowledge, data sharing, and collaboration among research institutions. It reflects the core values of open science such as quality, integrity, inclusiveness, and collective benefit, and sets guiding principles for the ethical and responsible conduct of publicly funded research. The updated legal framework aims to support the development of an enabling environment for open science practices across Mongolia. Unfortunately, it has not translated into English.

28. 2.3 In your country, are there specific policy instruments[i] that aim to promote open science?

[i] Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).

- ☐ Yes
- ☒ Under development
- ☐ No

29. 2.3 cont. **If yes or under development**, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

Mongolia has specific policy instruments aimed at promoting open science, primarily through administrative acts issued by relevant governmental bodies. These administrative acts provide technical and operational mechanisms to implement the goals set by national policies on science and technology. They focus on supporting open access to scientific publications and data, encouraging collaboration among research institutions, and ensuring transparency and accountability in publicly funded research. These instruments outline clear objectives, target beneficiaries, resources, and indicators to deliver tangible products and long-term impacts in advancing open science practices in Mongolia.

30. 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)

- ☐ Yes
- ☐ Under development
- ☒ No

31. 2.4 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

Mongolia does not currently have a formal, comprehensive national policy, strategy, or plan of action specifically dedicated to science, technology, and innovation (STI). However, the government and relevant ministries are actively working toward developing such a framework to better coordinate and support scientific research, technology development, and innovation across the country. Efforts are underway to consult stakeholders and align future policy initiatives with international standards and recommendations, including the principles outlined in the 2021 UNESCO Recommendation on Open Science.

32. 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: (ii) 17.d, 17.e, 17.g)

- ☒ Yes
- ☐ Under development
- ☐ No

33. 2.5 cont. **If yes**, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).

To view the elements of open science,

see: <https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/page=9>

In Mongolia, various research-performing institutions, universities, and scientific organizations are promoting open science in line with the 2021 UNESCO Recommendation. At the institutional level, universities such as the National University of Mongolia and the University of Finance and Economics, along with the Mongolian Academy of Sciences, have begun to implement internal strategies to encourage open access to research, data sharing, and broader collaboration. In addition, the Ministry of Economy and Development plays an active role in promoting open science by financing access to global open science resources and platforms. The Ministry currently covers subscription and access fees for major international databases and open-source platforms such as Springer Nature, EBSCO, and Dimensions. This support ensures that Mongolian researchers and students have equitable access to high-quality scientific knowledge, thereby reinforcing the values of inclusiveness, equity, and collective benefit outlined in the 2021 Recommendation. These efforts contribute significantly to building an enabling environment for open science across research institutions in Mongolia.

34. 2.6 In your country, are there specific funding mechanisms[i] for open science at the national level? (ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)

[i] A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.

- ☒ Yes
- ☐ Under development
- ☐ No

35. 2.6 cont. **If yes or under development**, please provide details and links as relevant.

While Mongolia does not yet have funding mechanisms specifically and exclusively dedicated to open science, the country does provide research grants through national funding programs that support publicly funded research. These grants, administered primarily through the Ministry of Education and Science and the Mongolian Foundation for Science and Technology, can be used to support activities that align with open science practices—such as open access publishing, data sharing, and collaborative research. Although not explicitly framed as “open science” funding, these mechanisms contribute indirectly to the advancement of open science by enabling research institutions and individual researchers to produce and disseminate scientific knowledge. The government also supports infrastructure development and capacity-building initiatives which can include the development of institutional repositories and research information systems. Expanding these mechanisms and integrating open science as an explicit funding objective remains a potential area for future policy development in Mongolia, in line with the principles outlined in the 2021 UNESCO Recommendation on Open Science.

36. 2.7 1.1 In your country, have open science practices[i] been integrated in existing research funding mechanisms?

[i] Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

☒ Yes

☐ No

37. 2.7 cont. **If yes**, please provide details and links as relevant.

In Mongolia, open science practices have been increasingly integrated into existing research funding mechanisms. National research grants—particularly those managed by the Ministry of Economy and Development and the Mongolian Foundation for Science and Technology—support activities that promote open access to scientific publications, research data sharing, and collaboration among researchers. These mechanisms also encourage engagement with societal stakeholders, including sectors such as health, environment, and agriculture, aligning with the principles of inclusiveness and collective benefit. In addition, Mongolia has initiated the development of a regulation for the establishment of an intersectional laboratory, which will serve as a shared platform accessible to researchers across various institutions and disciplines. This initiative is designed to foster interdisciplinary collaboration, promote the use of shared scientific infrastructure, and enhance the accessibility of research tools and data. It represents a concrete step toward enabling open science practices in line with the 2021 UNESCO Recommendation, particularly with respect to open infrastructure, researcher collaboration, and public engagement in science.

38. 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

☒ Yes

☐ No

39. 2.8 cont. **If yes**, please provide details and links as relevant.

Mongolia is actively working to foster equitable public-private partnerships (PPPs) in open science by the end of 2025, in alignment with the 2021 UNESCO Recommendation on Open Science. A significant development is the enactment of the Public-Private Partnership Law in December 2023. This law establishes a clear legal framework for implementing major infrastructure and public service projects through private investment on mutually beneficial terms. It aims to enhance coordination between public institutions and private entities, facilitating collaborative efforts in various sectors, including science and technology. Additionally, the Ministry of Economy and Development, in cooperation with the United Nations Conference on Trade and Development (UNCTAD), organized a national training on "Policies and Experiences of Science, Technology, and Innovation Parks" in April 2025. The training aimed to strengthen the foundation of development and the economy based on science, technology, and innovation in Mongolia. It brought together representatives of science and technology parks, business incubators, co-working offices, technology transfer centers, and startup entrepreneurs, who had the opportunity to meet and exchange experiences with delegates from countries that are successfully developing such park. Furthermore, the Mongolian National Chamber of Commerce and Industry (MNCCI) and the Science Park are collaborating to establish a working group, draft a cooperation agreement, and create a detailed plan. Their aim is to facilitate the exchange of scientific information and research findings between scientific organizations and businesses, thereby promoting innovation and technology transfer. These initiatives demonstrate Mongolia's commitment to developing a robust ecosystem for open science through equitable public-private partnerships, ensuring that scientific knowledge and innovations are accessible and beneficial to all sectors of society.

40. 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.j, 23.c)

☐ Yes

☐ Under development

☒ No

Part 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

41. 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

As of the latest data, Mongolia's research and development (R&D) expenditure as a percentage of Gross Domestic Product (GDP) was 0.10% in 2024, according to the Ministry of Economy and Development of Mongolia.

42. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

Individuals using the Internet (% of population) in Mongolia was reported at 73 % in 2023, according to the Statistics Department of Ulaanabaatar city. <http://ubstat.mn/StatTable.aspx?tableID=89>

43. 3.3 Are there national research and education networks (NRENs)[i] active in the country? (ref.: (iii) 18.c)

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

☒ Yes

☐ No

44. 3.3 cont. **If yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

Mongolia has an active National Research and Education Network (NREN) known as ErdemNet. Established in 2000, ErdemNet was created to enhance communication, information exchange, and research access within the fields of education, culture, and science across the country. In 2012, the network expanded with the establishment of an educational data center, supported by joint funding from the state budget and the Asian Development Bank. This expansion involved the procurement and installation of servers, equipment, and network resources to strengthen the infrastructure of the education sector's information system. By 2014, ErdemNet had taken over the internet services for schools and education departments in provincial centers, further solidifying its role as the backbone of Mongolia's NREN. ErdemNet's primary mission is to support the needs of the research and education communities in Mongolia by providing specialized internet services tailored to their requirements. This includes facilitating access to digital learning platforms, research databases, and collaborative tools essential for academic and scientific endeavors. In addition to ErdemNet, Mongolia has partnered with Eduten, a Finnish digital math learning platform developed by the University of Turku. In 2023, the Ministry of Education and Science of Mongolia signed a Memorandum of Understanding with Eduten to introduce this platform into Mongolian schools. The Eduten platform offers a library of over 200,000 pedagogically high-quality tasks for students aged 6 to 18, aiming to improve mathematics education in the country. These initiatives demonstrate Mongolia's commitment to enhancing its research and education infrastructure through the development and expansion of national networks and international collaborations.

45. 3.4 Are there national or institutional open science infrastructures[i] in your country? (ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

☒ Yes

☐ Under development

☐ No

46. 3.4 cont. **If yes or under development**, please indicate which type(s) of infrastructure:

- ☐ federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage
- ☐ community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society
- ☒ platforms for exchanges and co-creation of knowledge between scientists and society
- ☐ community-based monitoring and information systems to complement national, regional and global data and information systems

47. 3.4 cont. **If yes or under development**, for each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

48. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

- ☐ Yes
- ☒ No

49. 3.5. cont. **If no**, are the existing open science infrastructures at regional and international levels accessible to all the relevant open science actors from your country?

- ☐ Yes
- ☐ No
- ☐ Other

50. 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

- ☐ Yes
- ☒ No

Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

51. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

- ☒ Yes
- ☐ No

52. 4.1 cont. **If yes**, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

Mongolia has initiated the establishment of a new university under the Mongolian Academy of Sciences (MAS). On May 15, 2024, the Mongolian Cabinet approved a resolution to develop science and technology, which includes the founding of a university leveraging the MAS's existing infrastructure and human resources. The goal is to prepare specialized human resources in leading fields of science and technology, strengthen national capabilities, and intensify interdisciplinary research and development. The university will offer professional courses and programs in collaboration with relevant universities, focusing on technology transfer and the application of academic research into production and services. This initiative aims to enhance the integration of research and education, fostering innovation and the adoption of advanced technologies in Mongolia. The university is expected to play a pivotal role in developing a skilled workforce capable of addressing the country's scientific and technological challenges.

53. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

☒ Yes

☐ No

54. 4.2 cont. **If yes**, please provide more information and links as relevant.

Not yet systematically. In Mongolia, while general capacity-building efforts related to science, technology, and innovation (STI) have been implemented, there have not yet been formal or comprehensive training programmes specifically targeting policymakers on how to integrate the core values and principles of open science into STI policies and strategies. However, awareness is gradually increasing. Some indirect efforts have occurred through international cooperation, workshops, and participation in regional forums related to open science and digital transformation. These engagements have exposed policymakers to global best practices and the importance of openness, transparency, inclusivity, and collaboration in science policy. With the recent adoption of updated laws and strategic initiatives, such as the establishment of a university under the Mongolian Academy of Sciences and the modernization of digital platforms like E-Mongolia, there is growing momentum that could lead to more targeted capacity-building programs in the near future.

55. 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

☐ Yes

☒ No

56. 4.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

As of now, Mongolia has not yet systematically incorporated a comprehensive framework of open science competencies into higher education research curricula. However, there are ongoing initiatives that lay the groundwork for such integration. The Mongolian University of Science and Technology (MUST) has been a pioneer in promoting open education and digital learning. In 2019, MUST established the Open Education Center (OEC), focusing on developing Massive Open Online Courses (MOOCs), Open Educational Resources (OER), and blended learning models. The OEC has introduced training modules on topics such as Open Education, Instructional Design, and MOOCs, which indirectly support the development of open science competencies among educators and researchers. Additionally, in 2015, UNESCO and MUST launched an online course titled "New Technologies in Effective Teaching and Learning in Higher Education in the Context of Mongolia." This course, available in Mongolian, includes modules on Open Educational Resources, open licensing, and mobile learning, providing foundational knowledge that aligns with open science principles. While these initiatives are steps in the right direction, a fully integrated framework of open science competencies within higher education curricula has not been formally established. Looking ahead, the Education Sector Medium-Term Development Plan (2021–2030) envisions the transition of Mongolia's education system into a digital framework by 2030, which may include the development of such competencies. Therefore, while there is no formal framework in place yet, the foundational elements are being developed, and it is possible that a comprehensive integration of open science competencies into higher education curricula could be realized by the end of 2025.

57. 4.4 Are open educational resources[i] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

[i] Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

- ☐ Yes
- ☒ Partly
- ☐ No

58. 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

- ☐ Yes
- ☒ No

59. 4.5 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

60. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: (v) 20.b, 20c, 20i)

- ☒ Yes
- ☐ No

61. 5.1 cont. **If yes**, please provide more information and links as relevant.

Mongolia has initiated efforts to review and align research assessment and evaluation processes with open science principles, with further developments anticipated by the end of 2025. In April 2025, the Government of Mongolia approved new regulations—"Evaluation Procedures," "Monitoring Procedures," and "Evaluation and Review of Organizational Activities and Results"—under Government Resolution No. 43. These regulations aim to enhance the coherence of development policies, ensure regular measurement of implementation processes, and evaluate results based on research and analysis. The implementation of these procedures is expected to improve policy implementation, resource allocation, and accountability, aligning with open science values of transparency and evidence-based decision-making. Additionally, in November 2024, the Mongolian Academy of Sciences (MAS) co-hosted an international training workshop on open science infrastructures for disaster management. This workshop focused on building resilient open science infrastructures to support disaster management, emphasizing the importance of open data and collaborative research practices.

62. 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

☐ Yes

☒ No

63. 5.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

64. 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20.h)

☒ Yes

☐ No

65. 5.3 cont. **If yes**, for each initiative, please provide more information, including on the stakeholders[i] that are involved or lead the initiative, and links as relevant.

[i] For example, research funders, universities, research institutions, learned societies, scientific publishers and journal editorial boards.

Mongolia has initiated several measures to recognize and reward open science practices, with further developments anticipated by the end of 2025. In 2024, the 8th National Conference of Mongolian Young Scientists focused on "Open Science – Youth Participation," highlighting the importance of open science principles. The conference included the Khureltogoot Academic Conference, where over 500 research submissions were reviewed, and 28 studies received grants. Notably, the grant amount for young post-doctoral researchers was increased from 4 million MNT to 20 million MNT, reflecting a commitment to supporting open science initiatives. Additionally, the Innovation 7 Days event organized the Best Paper Awards, with a prize fund of 30 million MNT, recognizing outstanding research contributions. These awards provide incentives for researchers to engage in open science practices and disseminate their findings. Furthermore, the Russian Science Foundation and the Ministry of Education and Science of Mongolia (2024) signed a partnership agreement to support joint research projects. This collaboration encourages open sharing of research outcomes and fosters international cooperation, aligning with open science principles.

66. 5.4 Have there been any unintended negative consequences[i] of open science practices reported in your country? (ref.: (v) 20)

[i] For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

☐ Yes

☒ No

Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES

OF THE SCIENTIFIC PROCESS

67. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: (vi) 21)

- ☒ YES, at the national level
- ☐ YES, at the institutional level
- ☐ YES, at both the national and institutional levels
- ☐ No

68. 6.1 cont. **If yes**, please indicate which of the recommended actions they address:

- ☐ promotion of pre-prints
- ☐ exploration of open peer-review practices
- ☐ valuing and sharing of negative scientific results and associated data
- ☐ participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- ☒ development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas
- ☒ promoting open innovation practices
- ☐ Other

69. 6.1. cont. **If yes**, please provide more information and links as relevant

Part 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

70. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

Mongolia is actively promoting and facilitating international scientific collaboration in alignment with the 2021 UNESCO Recommendation on Open Science. Through bilateral and multilateral agreements, Mongolian research institutions collaborate with partners from countries such as Japan, South Korea, China, Vietnam and Russia, particularly in areas of mutual interest like sustainable development, environmental science, digital innovation, and climate change. Mongolia also participates in regional and global scientific initiatives under the auspices of UNESCO and other international organizations, enabling the exchange of knowledge, data, and best practices.

71. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

- ☐ Yes
- ☒ No

72. 7.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

73. 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: (vii) 22.c)

☐ Yes

☒ No

74. 7.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Part 8. GENERAL CONSIDERATIONS

75. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

☐ Yes

☒ No

76. ANY OTHER COMMENTS: If you wish, you may enter any additional comments or information here. You may also contact openscience@unesco.org